```python
# Logistic Regression Implementation in Python

import numpy as np
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import confusion_matrix, accuracy_score, classification_report

# 1. Load dataset
iris = load_iris()
X = iris.data
y = iris.target

# 2. Split dataset into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# 3. Initialize Logistic Regression model
# 'max_iter' ensures convergence
model = LogisticRegression(max_iter=200)

# 4. Train the model
model.fit(X_train, y_train)

# 5. Predict on test data
y_pred = model.predict(X_test)

# 6. Evaluate performance
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))
print("\nAccuracy:", accuracy_score(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred, target_names=iris.target_names))

# 7. Predict a new sample
new_sample = np.array([[5.1, 3.5, 1.4, 0.2]])  # Example flower
prediction = model.predict(new_sample)
print("Predicted Class for new sample:", iris.target_names[prediction][0])
```

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Confusion Matrix:
 [[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]

Accuracy: 1.0

Classification Report:
               precision    recall  f1-score   support

      setosa       1.00      1.00      1.00        10
  versicolor       1.00      1.00      1.00         9
   virginica       1.00      1.00      1.00        11

    accuracy                           1.00        30
   macro avg       1.00      1.00      1.00        30
weighted avg       1.00      1.00      1.00        30

Predicted Class for new sample: setosa
```